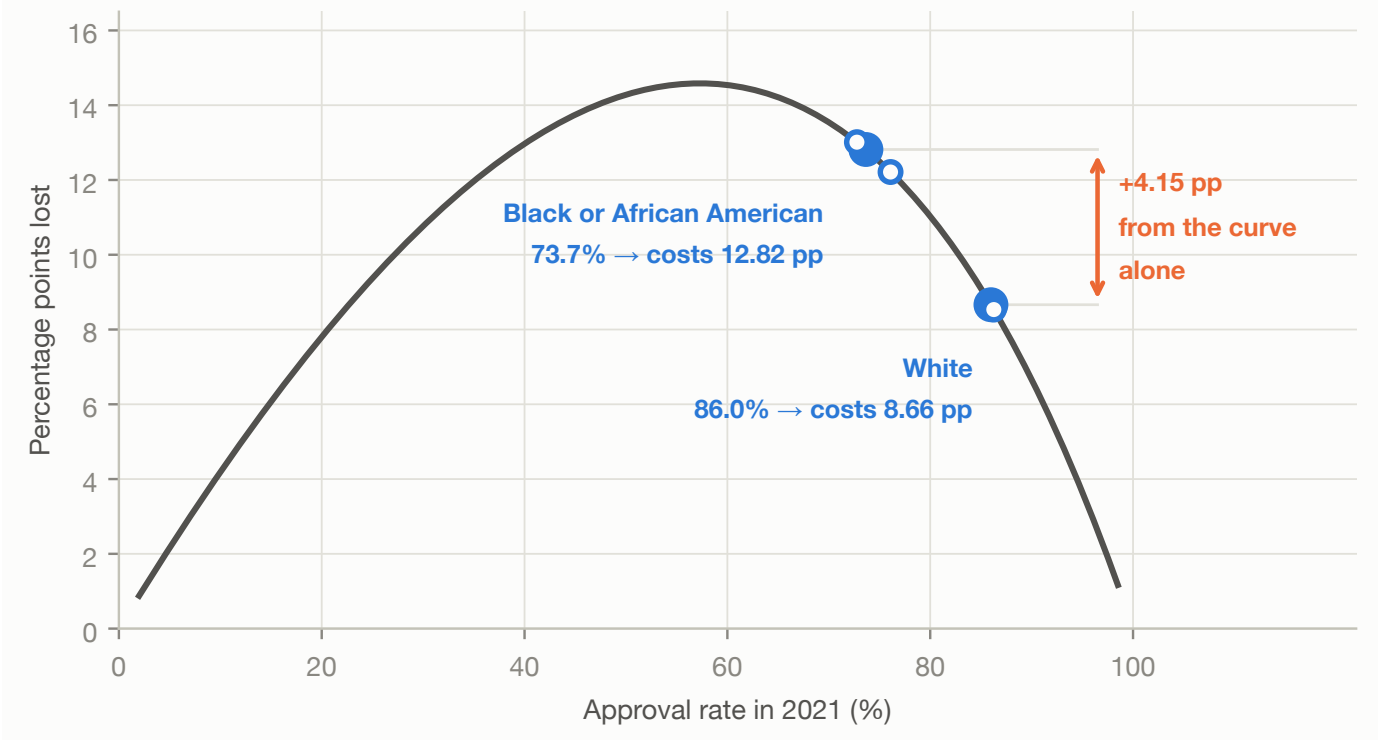


Widening by Curvature

Between 2021 and 2023 the mortgage approval gap between White and Black applicants widened by **3.26 points**.

In odds, Black applicants tightened **less** than White ones. Both figures are correct.



+3.26 pp observed widening, White vs Black	+4.15 pp what a group-blind odds shift produces	0.576 / 0.554 odds ratio, Black / White
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Widening by Curvature: What a Percentage-Point Gap Between Groups Measures, HMDA 2021–2023

A gap measured in percentage points widens when a single group-blind tightening is applied to groups that start from different places. The arithmetic accounts for most of the widening observed in the 2021–2023 mortgage contraction — which licenses no conclusion about treatment in either direction.

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Working paper · September 2026 · Data and code: github.com/cperezrisquet/approval-rate-reversal

ABSTRACT

Between 2021 and 2023 the pooled approval rate on mortgage applications reported under the Home Mortgage Disclosure Act fell 9.38 percentage points, and the approval gap between groups widened on every axis examined: 3.26 pp between White and Black applicants, 4.43 pp between White and American Indian or Alaska Native applicants, 3.15 pp by ethnicity, 1.03 pp by sex. Applying a *single group-blind odds ratio*, estimated across all strata of each axis, to the 2021 baselines reproduces between 46 % and 128 % of each of those widenings, with absolute residuals between 0.09 and 0.90 pp. The mechanism is the curvature of the logistic scale: the same odds shift costs $r(1-r)(1-\psi)/(1-r+\psi r)$ percentage points, which is zero at both ends and maximal in the middle, so groups starting further from 100 % lose more points from identical treatment. Measured in odds instead, the tightening is not monotone in anything: Native Hawaiian or Other Pacific Islander applicants tightened 15 % more than White applicants and Black applicants 4 % less. The paper argues that a percentage-point gap is the wrong instrument for monitoring differential outcomes, and states explicitly that neither the widening nor its arithmetic explanation supports an inference about differential treatment, in either direction.

1. Introduction

A gap between two groups' rates, quoted in percentage points, is the standard instrument for monitoring differential outcomes. It is on fair-lending dashboards, in regulatory filings, and in the press. It has the virtue of being immediately interpretable: a gap of twelve points means twelve applicants in a hundred.

It also has a property that is rarely stated where it is used: it moves when relative treatment does not. Lenders do not tighten by subtracting percentage points; they move a score cutoff, a debt-to-income ceiling, a loan-to-value floor. That is a multiplicative shift in odds, and a multiplicative shift in odds produces a percentage-point loss that depends on where each group started. Groups near the ceiling lose few points; groups in the middle lose many. The gap between them widens with no change in how either is treated.

This is not a new observation about effect measures. The choice between difference and ratio scales, and the fact that interaction is scale-dependent, is old and settled in epidemiology (Sheps, 1958; Greenland, 1987; Rothman, Greenland & Lash, 2008). What this paper adds is the measurement: how much of an actual, widely-noted widening in mortgage approval gaps is accounted for by curvature alone, under a counterfactual constructed so that it cannot be tuned to the answer.

The 2021–2023 contraction is a good test case because the tightening was large, abrupt and system-wide. The pooled approval rate fell 9.38 pp, of which an exact decomposition attributes -7.61 pp to within-stratum tightening and -1.77 pp to a change in the mix of products applied for. It is the within-stratum term that this paper is about.

2. Data and method

All figures come from filings under the Home Mortgage Disclosure Act, retrieved through the public Data Browser API of the Federal Financial Institutions Examination Council. No key or registration is required. The unit of observation is a loan application; the analysis uses the aggregated cell counts the API returns.

Two properties of that endpoint shape the design. It does not serve a national aggregate, so a geographic filter is mandatory; the 50 states and the District of Columbia are requested, and that set defines the population. A multi-valued filter also acts as a grouping axis, so requesting the five race categories returns the race-by-outcome cross-tabulation directly. The endpoint does *not* expose income, loan-to-value, debt-to-income or applicant age as grouping axes — this was tested; it accepts them and returns an undifferentiated total. Those fields exist only in the loan-level files. Section 8 states what that costs the analysis.

2.1 The approval rate is a definition, not a field

HMDA reports `action_taken`, not an approval rate. The rate used throughout is codes 1 (originated) and 2 (approved but not accepted) over codes 1, 2 and 3 (denied), excluding 4 (withdrawn), 5 (closed for incompleteness) and 6 (purchased loan). Codes 4 to 6 are excluded because the rate should measure lender decisions and a withdrawal is not one. Including withdrawals lowers the level by roughly 16 points and is common in practice; it does not affect any comparison reported here, because every comparison is between groups measured the same way.

2.2 Estimands

For a group with approval rate r , the odds are $o = r/(1-r)$. A tightening that multiplies odds by $\psi < 1$ produces a new rate

$$r' = \psi o / (1 + \psi o) = \psi r / (1 - r + \psi r)$$

and therefore a percentage-point loss of

$$r - r' = r(1-r)(1-\psi) / (1 - r + \psi r)$$

The factor $r(1-r)$ is the whole argument of this paper: it vanishes at $r = 0$ and $r = 1$ and peaks in between, so the cost of an identical shift is a function of the starting point.

The common shift ψ is estimated by Mantel–Haenszel (1959) across *all* strata of an axis, never across the pair being examined, so the counterfactual cannot be fitted to the gap it is meant to explain. Stratum-specific odds ratios carry Woolf intervals on the log scale. Whether the strata share a common shift at all is tested with Breslow–Day (Breslow & Day, 1980); section 6.1 is about why that test's answer should not be used to rank axes.

2.3 Pre-registration, and the hypothesis that failed

STATED BEFORE THE DATA WERE EXAMINED

The mechanism above generates a prediction: because the curve is steeper in the middle, the percentage-point loss should be largest for strata with the lowest baseline. Pooled across the 28 strata of all eight axes, that correlation is $r = -0.025$ — absent. The prediction was **false**, and the test design was also wrong: pooling strata from incommensurable axes lets opposing patterns cancel. Government-backed programmes barely tightened (VA -2.51 pp, RHS/FSA -2.97 , FHA -3.66) while conventional lending took -10.66 , an ordering that has nothing to do with baseline position. The curvature effect appears *within* the demographic axes and is a second-order effect overall. The record is in the repository, committed before the results.

3. Result 1: every gap widened

Measured in percentage points, the approval gap widened on all four large-stratum pairs examined. The pairs were fixed in advance and all four are reported.

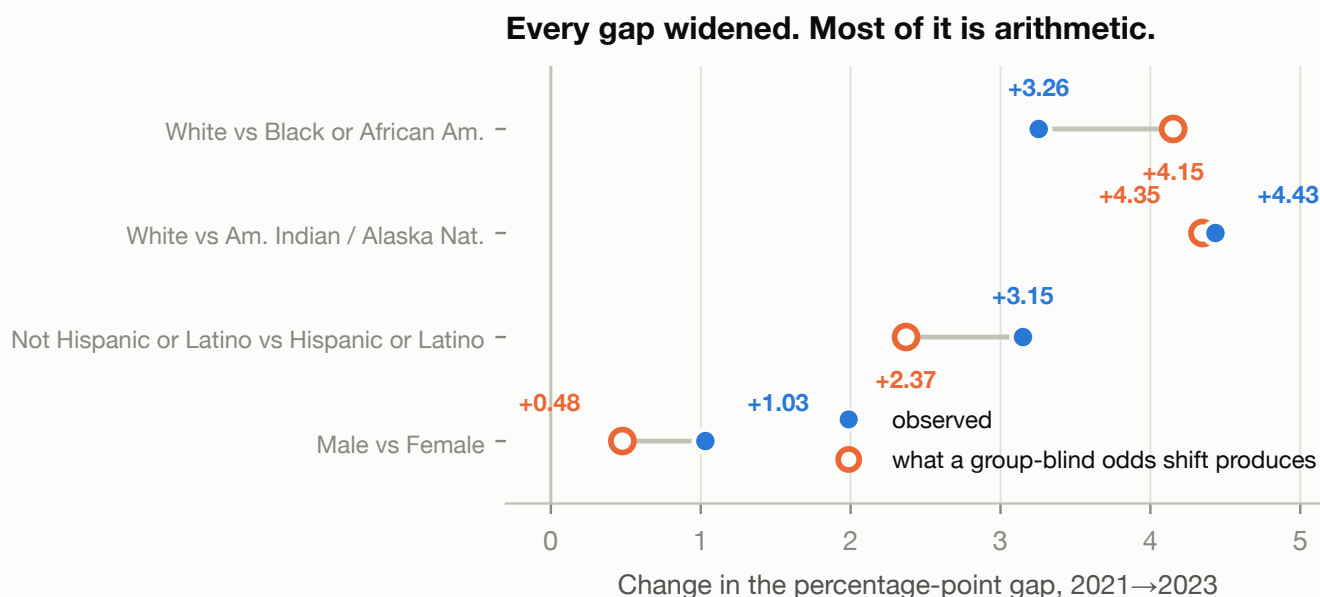


Figure 1. Change in the percentage-point approval gap, 2021–2023, for the four large-stratum pairs. The hollow marker is what a single group-blind odds shift applied to the 2021 baselines produces; ψ is estimated by Mantel–Haenszel across all strata of the axis, never across the pair, so the counterfactual cannot be fitted to the gap it explains. It over-predicts on White versus Black and under-predicts on the other three.

Table 1. Approval-rate gaps in percentage points, 2021 and 2023, with the widening produced by applying a single group-blind odds shift to the 2021 baselines. ψ is estimated across all strata of the axis, not across the pair.

Pair	Gap 2021	Gap 2023	Observed Δ	ψ	Δ under one blind shift	Residual
White vs Black or African American	12.33	15.58	+3.26	0.5557	+4.15	−0.90
White vs American Indian or Alaska Native	13.22	17.65	+4.43	0.5557	+4.35	+0.09
Not Hispanic or Latino vs Hispanic or Latino	6.13	9.28	+3.15	0.5516	+2.37	+0.78
Male vs Female	1.24	2.27	+1.03	0.5477	+0.48	+0.55

Read on this scale alone, the conclusion is straightforward: the contraction fell unevenly, and a breakdown by group detects it. Every gap moved in the same direction, and the largest movements are on the axes where the baseline levels are furthest apart.

4. Result 2: one group-blind shift accounts for almost all of it

The counterfactual asks what the gaps would have done had every stratum been subject to the same multiplicative tightening. It reproduces 46 % to 128 % of each observed widening, with absolute residuals between 0.09 and 0.90 pp against widenings of 1.03 to 4.43 pp.

For White versus Black the counterfactual *over*-predicts: a blind shift produces +4.15 pp where +3.26 pp was observed. That result is not an artefact of estimating ψ . Its 95 % interval is [0.5544, 0.5570], across which the

predicted widening moves only between +4.14 and +4.17 pp. The over-prediction of 0.90 pp survives the whole interval. It is specific to this pair and is not generalised: the other three pairs are under-predicted.

5. Result 3: in odds, the tightening is not monotone in anything

The same data on the multiplicative scale, with intervals, ordered by the estimate.

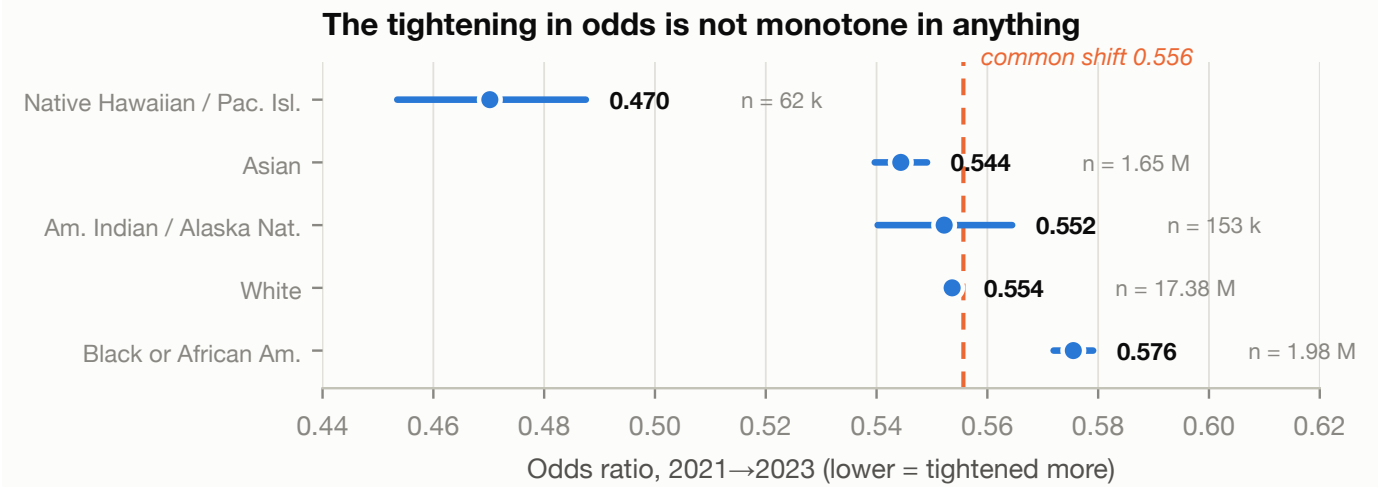


Figure 2. The same tightening on the multiplicative scale: odds ratio 2021→2023 by reported race, with 95 % Woolf intervals, ordered by the estimate. Lower means tightened more. The dashed line is the common shift of 0.5557 that Figure 1 uses. No monotone description of these five estimates survives, which is why the paper reports all of them rather than a comparison.

Native Hawaiian or Other Pacific Islander applicants tightened 15.1 % more than White applicants (odds ratio 0.4702 against 0.5536), the largest deviation in the table. Asian applicants tightened 1.7 % more. American Indian or Alaska Native applicants are indistinguishable from White ($p = 0.82$). Black applicants tightened 4.0 % less, and the intervals do not overlap.

No monotone description of these five numbers survives, and the paper does not attempt one. Reporting the Black–White comparison as the finding, with an odds ratio of 0.4702 sitting in the same column, would be selecting the comparison that suits a narrative — which is the error this series exists to document. All five strata are reported, ordered by the estimate.

6. Result 4: the curvature that produces the widening

The two scales disagree because they are different functions of the same change. Under a shift of $\psi = 0.5557$, the percentage-point cost peaks at a baseline of 57.2 % and falls away toward both ends.

What one group-blind odds shift costs, by starting point

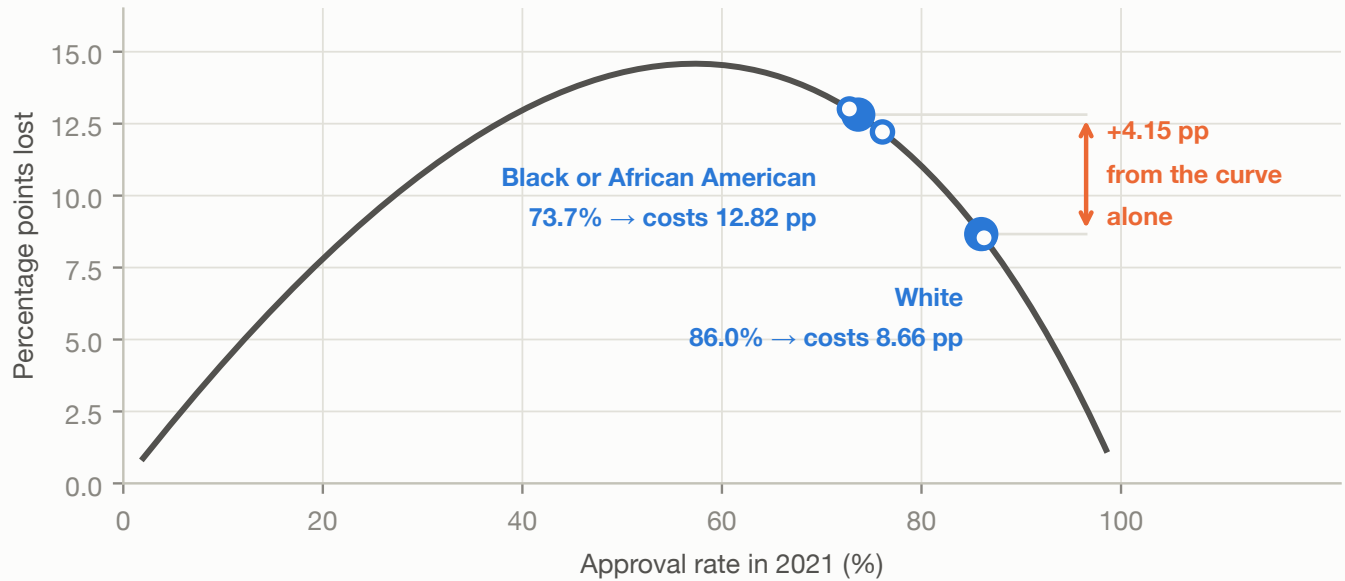


Figure 3. The mechanism. Percentage points lost to a single odds shift of $\psi = 0.5557$, as a function of the rate a group starts from — the function $r(1-r)(1-\psi)/(1-r+\psi r)$, zero at both ends and maximal at 57.2 %. Markers are the five race strata at their 2021 baselines. White starts at 86.0 % and loses 8.66 pp; Black or African American starts at 73.7 % and loses 12.82 pp. The 4.15 pp between them requires no group to be treated differently.

Table 2. Percentage-point cost of a single odds shift of $\psi = 0.5557$, by starting rate. The column is $r(1-r)(1-\psi)/(1-r+\psi r)$.

Baseline	After the shift	Points lost
50 %	35.72 %	14.28
60 %	45.46 %	14.54
70 %	56.46 %	13.54
80 %	68.97 %	11.03
86 %	77.34 %	8.66
90 %	83.34 %	6.66
95 %	91.35 %	3.65

White applicants started 2021 at 85.99 % and an identical shift costs them 8.66 pp. Black applicants started at 73.66 % and the same shift costs 12.82 pp. The difference of 4.15 pp is the gap widening the counterfactual predicts, and it arises with no group being treated differently from any other.

6.1 The test inverts the ranking it is asked for

An institution deciding which breakdowns deserve monitoring might reasonably test whether each axis's strata share a common shift, and attend to the axes that fail. Breslow–Day, applied to the three demographic axes, ranks sex as the most heterogeneous by a wide margin. The magnitudes rank it the least.

The test and the magnitude disagree on which axis matters

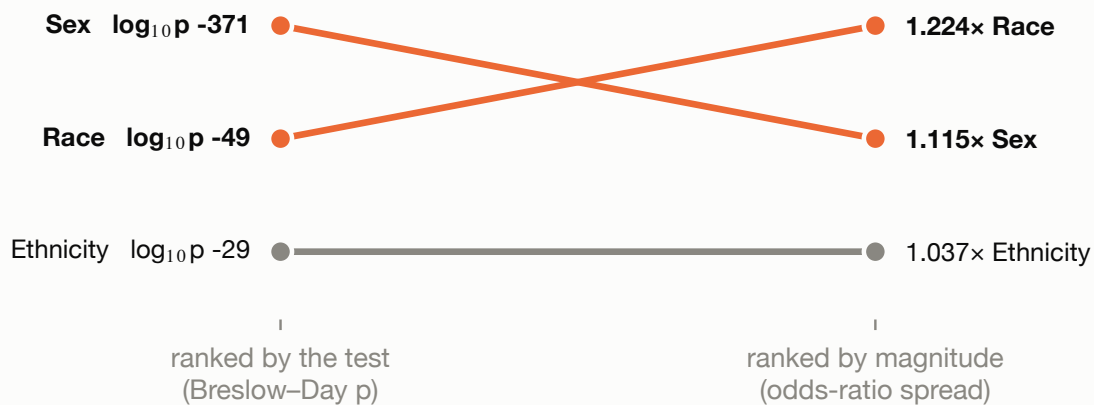


Figure 4. The same three demographic axes ranked two ways. On the left, by the Breslow–Day test of whether their strata share a common shift; on the right, by the magnitude of the departure from one. Sex leads the test by 321 orders of magnitude and trails the magnitude. The ranking the test produces is driven by sample size and degrees of freedom, not by how unequally the tightening landed.

Sex returns $\log_{10} p = -370.9$ with an odds-ratio spread of 1.115x; race returns $\log_{10} p = -48.8$ with a spread of 1.224x. The test places sex 321 orders of magnitude ahead of race on a question where the magnitude places it behind. The reason is arithmetic rather than substantive: sex carries 24.1 million applications over 2 degrees of freedom, race 21.2 million over 4. At these volumes the test's answer is largely a statement about the volume (Berkson, 1938; Lin, Lucas & Shmueli, 2013), which is why every quantity in this paper is reported as a magnitude with an interval.

7. Discussion

The measurement claim is narrow and, the paper argues, secure: a percentage-point gap between groups is a poor instrument for monitoring differential outcomes, because it responds to the level each group starts from as well as to how each is treated. Two institutions with identical policies and different customer mixes will show different gap dynamics. The same institution will show a widening gap in a downturn and a narrowing one in a recovery with its credit box untouched. An odds ratio with an interval does not have this property, is what the underlying decision rule actually shifts, and is what this paper recommends publishing alongside — not instead of — the levels themselves.

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Not this: that the observed widening demonstrates increased differential treatment. Most of it is reproduced by a shift that is blind to group by construction, and on the largest pair the observed widening is *smaller* than the blind counterfactual.

And not this: that lending was therefore equitable. This design has no applicant-level controls for income, collateral, debt burden or credit history, so it cannot compare like with like, and it says nothing about the level of any gap — only about how a gap *moves* when a system tightens. The 2021 gaps themselves, 12.33 pp between White and Black applicants before any of this began, are outside what is analysed here and are not explained by curvature.

Both inferences are drawn in practice, from the same correct figures. That is the subject of this series.

8. Limitations

The decisive limitation is the absence of applicant-level solvency controls. The aggregations endpoint does not expose income, loan-to-value, debt-to-income or age as grouping axes, and HMDA does not publish credit scores at all. Nothing here separates a change in who applied within a group from a change in how that group was treated, and no causal reading is available. A version with the loan-level files, several gigabytes per year, could stratify on the credit variables and would be a materially stronger design.

The counterfactual assumes a single multiplicative shift is the right description of an aggregate tightening across thousands of institutions. It is a benchmark, not a model of anyone's underwriting; its purpose is to establish how much of a widening needs no differential treatment to occur, not to claim that this is what happened. Breslow–Day rejects a common shift on all three demographic axes, so the strata demonstrably do not share one exactly — the point of section 6.1 is that the magnitude of that departure is small, not that it is zero.

The population is mortgage applications in 50 states and the District of Columbia, an unusually collateralised and well-instrumented segment. The demographic fields are as reported by the filing institution, with substantial non-response, and the strata are the API's own categories, which collapse distinctions that matter. Native Hawaiian or Other Pacific Islander carries 61,746 applications across both years, and its interval is correspondingly wide. Applicants reporting multiple races, and the API's joint and not-available categories, are outside the five strata used.

Finally, 2021 and 2023 are endpoints chosen because they bracket the contraction. The pooled rate was near its maximum in 2021, which is where curvature has the most room to act; a pair of years in a flatter part of the cycle would show smaller mechanical effects.

9. Reproducibility

All series, analysis code and figure generation are available at github.com/cperezrisquet/approval-rate-reversal. The API responses that feed the analysis are committed with it — they are public aggregate counts of roughly 500 bytes each — so every number here can be rechecked without contacting the API. The Mantel–Haenszel estimator is checked against `statsmodels`; Breslow–Day p-values below the floating-point floor are computed from the asymptotic expansion of the upper incomplete gamma function and checked against `scipy` over the range where `scipy` does not underflow. The pre-registration and the failed hypothesis are in `findings/`, committed before the results they concern. Data retrieved 10 September 2026, covering HMDA filing years 2018 through 2025.

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Prepared from public sources. HMDA data is published by the Federal Financial Institutions Examination Council and is cited here as a data provider. This paper analyses how a between-group gap responds to an aggregate tightening; it has no applicant-level controls and makes no finding about lending discrimination, in either direction. Section 7 states the two inferences it does not license. Filings are subject to revision by the source. · Dr. Sc. Carlos Pérez Risquet, September 2026